

MSc by Research Integrative Neuroscience

Project proforma

Primary Supervisor

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Secondary supervisor (if relevant)

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Location

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Town	Edinburgh

Expert Marker (Project 1 research proposal; Project 2 project dissertation)

Please give details of an expert marker who has agreed in principle to mark the assessments related to the project. Details will be passed to the Board of Examiners for use if appropriate.

Name	Dr Matthew Iveson
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Project Details

Project Title or theme	Does childhood adversity have lifelong consequences for circadian and sleep health?
Brief summary (max 200-300 words)	Background Early life adversity has been associated with lifelong effects on stress reactivity and a variety of negative neuropsychiatric outcomes including mood and anxiety disorders (Carr <i>et al.</i>, 2013) and insomnia (Gregory <i>et al.</i>, 2006). These effects of childhood adversity are thought to originate from early life alterations in the HPA axis that cause lifelong vulnerability to stress through orexin and noradrenergic systems, hippocampal neurogenesis and epigenetic mechanisms. (Palagini <i>et al.</i>, 2015). Previous studies have reported associations between childhood adversity and poor sleep at all stages of later life. While sleep and circadian disruption are commonly associated with trauma exposure in adulthood, they have been under-investigated in individuals with experience of early life adversity. Sleep and circadian disruption may therefore be under-appreciated mediators of the enduring impact of childhood adversity on vulnerability to poor physical and mental health. To the best of our knowledge, links between childhood adversity and adult sleep have not been investigated in the UK population. Furthermore, there have been no previous studies of the effects of early life adversity on circadian phenotypes (chronotype) in humans or in animal models. In the proposed study, we address these gaps in the literature in UK Biobank, by performing the biggest (~150,000 participants) population-based study of

	childhood adversity, sleep and chronotype to date. The extensive lifestyle, health, demographic and genetic data available in UK Biobank will allow us to investigate the impact of multiple potential demographic and lifestyle-related confounding factors, including genetic predisposition to unfavourable sleep phenotypes. To investigate the impact of early life adversity on sleep and circadian profile, we will use comparator groups of people with experience of adulthood adversity and individuals with no experience of adversity (adulthood or childhood). We will additionally investigate whether sex is a significant contributing factor to the extent of sleep or circadian changes. The results of the proposed study will demonstrate whether adverse events in childhood affect sleep decades later. Our findings will further understanding of the enduring effects of childhood adversity on health and wellbeing and support consideration of childhood adversity in the prevention and intervention of sleep disruption.
Techniques	Analysis of cross-sectional cohort data (UK Biobank) using R
Reading (2-3 references)	- Chapman, Daniel P., Anne G. Wheaton, Robert F. Anda, Janet B. Croft, Valerie J. Edwards, Yong Liu, Stephanie L. Sturgis, and Geraldine S. Perry. "Adverse Childhood Experiences and Sleep Disturbances in Adults." <i>Sleep Medicine</i> 12, no. 8 (September 2011): 773–79. - Fuligni AJ, Chiang JJ, Tottenham N. Sleep disturbance and the long-term impact of early adversity. <i>Neurosci Biobehav Rev.</i> 2021 Jul;126:304-313. doi: 10.1016/j.neubiorev.2021.03.021. Epub 2021 Mar 20. PMID: 33757816. - Yapp E, Booth T, Davis K, Coleman J, Howard LM, Breen G, Hatch SL, Hotopf M, Oram S. Sex differences in experiences of multiple traumas and mental health problems in the UK Biobank cohort. <i>Soc Psychiatry Psychiatr Epidemiol.</i> 2023 Dec;58(12):1819-1831. doi: 10.1007/s00127-021-02092-y. Epub 2021 May 10. PMID: 33970300; PMCID: PMC10628045.
Type	Analysis project (student undertakes analysis of existing data and/or meta-analysis and/or systematic review of the literature)

Information

# of spaces on this project	1
Meetings to discuss project	Weekly: Wednesdays at 10am (in-person or online as necessary)
Is a Home Office Licence personal licence required?	No
Does this project require access to NHS data or patient TRAK records that limits its availability to students with appropriate clearance?	No
Does this project require contact with NHS patients?	No
Does this project require Hepatitis B Immunisation?	No
Does this project have ethical approval in place?	Yes
Are the techniques involved in the project regularly in use in your laboratory?	Yes
If the answer to the above is 'NO', have the techniques involved been tested and shown to work in your laboratory?	
Notes	